

FPGA Enactment of Biological Feature Based Authentication Algorithm Using Elliptic Curve Cryptography for IoT Communication

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Abstract—In the modern medical environment, the Internet of Things-enabled devices helps physicians to control and remotely monitor the device involved in the healthcare sector. Here Biological features such as DNA strands are best suitable to store and share a large number of patient's sensitive data in encrypted form and also help to provide a higher level of security. Traditional Rivest Shamir Adleman cryptographic-based authentication algorithms have security problems and cannot able to hold a large amount of data, also require more memory, latency, and computational speed. To overcome the above limitations, biological feature-based lightweight Elliptic Curve Cryptography-based authentication is developed for resource-constrained IoT communication. The proposed ECC-based authentication and integrity checking architecture consists of internal error control coding technique. It generates the DNA code and appends it with an encrypted message to strengthen the authentication process. Here ECC is used to encrypt the original plaintext, achieves greater security to authenticate IoT devices, and consumes fewer resources such as memory, area, and power.

Keywords—*elliptic curve cryptography, biological feature, DNA cryptography, authentication algorithm, internet of thing, internal error control, external error control.*

I. INTRODUCTION

In modern medical field, it is very essential to safeguard the patient's identity which leads to fraudulent use of the person's credit cards, loans, and other financial accounts. Also, Medical health record data can be used to commit insurance fraud. This could result in the patient being denied coverage or having to pay higher premiums. In some cases, medical health record data could be used to harm a person physically. For example, if a person's allergies were compromised, they could be exposed to an allergen that could cause a serious reaction. For these reasons, it is crucial that medical health record data need to be secure while transferring using Medical IoT technology. There are a number of steps that can be taken to secure this data, which includes lightweight cryptographic algorithm to convert the patient data into a format that cannot be read without a special key. This makes it very difficult for unauthorized individuals to access the patient's data. Another way is to restricting access to the data by allowing only authorized doctor to should have access to patient medical health record. This can be accomplished by setting access control policies that limit who has access to the data and what they can do with it [7][12].

Many different conventional encryption methods, such as AES, DES, and RSA, can be employed to secure medical health record data [8]. But these methods are not suitable for the resource constrained Medical IoT devices. So the

bioinspired and lightweight cryptographic authentication algorithm is proposed to secure the patients' health record by involving Elliptic Curve Cryptography to safeguard the data and also encode the data using DNA code to provide the double level of encryption. The proposed authentication algorithm also implies the control policies internally and externally by implementing two-factor authentication. Here, Elliptic Curve Cryptography is used to encrypt the patients' health due to its smaller key size which is more suitable for resource constrained Medical IoT platform [9][11]. It makes ECC is faster than Rivert Shamir Adleman (RSA) Algorithm and more resilient to attack than RSA. The main aim of the proposed authentication algorithm is to authenticate the medical devices such as pacemaker and insulin pump of the patients which helps to ensure that only authorized doctor to access patient data and Devices [10]. It also helps to protect integrity of the medical data during transmission. The main contribution of the author is mentioned as follows:

- A novel DNA code generation method is introduced as Message Authentication Code (MAC) to perform secure data integrity check.
- A lightweight ECC encryption and decryption algorithm is used to improve the confidentiality and security level with less computational complexity.
- Internal and External Error Control Code bit-based authentication algorithm is proposed to achieve better security level.

The paper is organized in such a way that to achieve the above-mentioned lightweight authentication algorithm. To begin with introduction about brief history of Elliptic Curve Cryptography, DNA cryptography and Message Authentication Code. Section 2 illustrates the related work. Section 3 gives the detailed information about biological feature-based authentication algorithm. Section 4 deliberates about the implemented results and finally concludes the paper.

II. RELATED WORKS

With smart gadgets becoming easier to get, the Internet of Things (IoT) has become an important way to share information and communicate. Mutual authentication and session key negotiation are necessary to make sure that IoT networks and faraway users can send and receive data safely. This study suggests an ECC-based three-factor remote user authentication scheme that protects privacy and data confidentiality. The plan can't be broken with cryptographic attacks, and it's not as heavy as other methods. Using the AVISPA simulation tool to do a formal security analysis shows that it is strong against security threats. The proposed

scheme takes care of security problems like the privacy, integrity, and authentication of data while keeping computation and communication costs as low as possible. Informal security analysis and formal AVISPA simulation show that the method is secure. When compared to other plans like it, the results show that it is more secure and works better. The scheme has stages for updating symmetric keys and passwords, so it can be used in real life. Sadhukhan et al. [1] show how a mixed cryptosystem made up of Elliptic Curve Cryptography (ECC), Advanced Encryption Standard (AES), and a one-way hash function can be used to make cloud storage safe. This safe way to store data is important for things like military databases, medical imaging systems, and online picture albums. The suggested system is better in terms of space complexity than RSA and CRT, and ant colony optimization makes it better at finding the best solution. The system is built on the platform Azure and tested in NS2. The results show that the proposed HECC is the best in terms of how long it takes to trade keys, how much space it takes up, and how much energy it uses. This method is used successfully in the suggested military application, which means that no packets are lost and response times are faster.

Yao, Hailong, et al. [2] show the structural flaws and security holes in current schemes and suggest a new ECC-based multi-factor remote authentication and access control scheme for E-healthcare. The new scheme improves usefulness and security while keeping the amount of work needed to do computations and communicate at a low level. The proposed scheme offers mutual authentication, access control, session key security, anonymity, and forward security. It can also stop known Internet attacks like insider attacks, attacks that use multiple factors of security, attacks that pretend to be someone else and attacks that use intermediate data. The suggested scheme has a total communication overhead of only 520 bytes, and the communication overhead on the user side is almost the same as other schemes. Due to improvements in wireless communication technology, RFID authentication methods are becoming more and more important in healthcare settings. The security of Liao and Hsiao's suggested protocol comes from the elliptic curve cryptosystem (ECC), but it has a problem called "key compromise." Zhao et al. [3] suggest a new ECC-based procedure that gets around these problems and works just as well, making it better for use in healthcare settings. The proposed protocol is faster and safer than the earlier ECC-based protocols. This makes it a better choice for use in healthcare settings.

Sowjanya, K., et al., [4] suggest an improved lightweight ECC-based end-to-end authentication system that gets rid of these problems and makes things simpler overall. The storage overhead is cut by 68.57 percent, the communication overhead is cut by 37.93 percent, and the computing overhead is cut by more than that because of the extra ECC scalar point multiplication operations. BAN logic and the AVISPA tool are used to do both formal and informal security research. The suggested authentication protocol is better than similar plans and can be used in the real world. Many IoT devices don't have enough memory or processing power to handle asymmetrical cryptography functions. To fix this, programmers must come up with their own security protocols or use stripped-down versions of protocols that are already in use. V. Kamalakannan and Tamilselvan S's paper [5] is about a high-performance Elliptic Curve Cryptoprocessor made for Field Programmable Gate Array (FPGA) implementation using asymmetric and

hash methods. For encryption and decryption, a new cryptographic method and the Elliptic Curve Digital Signature method (ECDSA) are used. The suggested cryptoprocessor has a minimum period of 6.980 ns and a maximum frequency of 143.276 MHz. This makes it possible for devices with limited resources to use asymmetrical cryptography.

Khan, Mohammad Ayoub et al. [6] proposed the secure framework to authenticate the patient data by using patient's biometric information as password and username. Improved ECC and Substitution ceaser cipher techniques are used to send the patient data securely from client to server. A new secret key is generated by using existing private key, base point and public key of the user. The generated secret key is used for ECC encryption and decryption algorithm to send the data securely to the cloud storage server. There are so many Fuzzy based ECC encryption and decryption is proposed for different application [13][14][15]. But the proposed secure framework was implemented and proved that it achieved very less computational cost and communication cost compared with the other existing work.

III. PROPOSED BIOLOGICAL FEATURE BASED LIGHTWEIGHT ECC BASED AUTHENTICATION ALGORITHM FOR IOT COMMUNICATION

A lightweight biological feature-based authentication algorithm is proposed for IoT communication. The proposed authentication algorithm was developed by using an internal error control coding and external error control coding technique which strengthens the security level. Both error control code-based authentication algorithm provides difficulty to eavesdroppers to generate the ciphertext. Because the decryption algorithm needs a valid error control bit to generate the plaintext, even an eavesdropper will have the DNA code. A novel Ex-OR operation-based DNA code generation method is used to generate the code and it plays a vital role in message integrity and the authentication process. Elliptic Curve Cryptography is used for encryption and decryption algorithm to maintain the confidentiality of the user and it uses only 160-bit key size for cryptosystem process. Before that, plaintext needs to be converted into plaintext point by using the koblitz method. The following steps are used to encode the message into DNA code by using XoR operation.

Step 1 : Choose the plaintext (m)

Step 2 : Convert the plaintext into ASCII code by using Table 1

Step 3 : Transform the ASCII code into plaintext point by using Koblitz method.

Step 4 : Translate the plaintext point (Pm) into binary digit (Bm).

Step 5 : Divide binary digit blocks into four equal sub-blocks, namely M1, M2, M3, and M4.

Step 6 : Perform the Ex-OR operation between two sub-blocks M1, M2 and M3, M4.

$$M12 = M1 \oplus M2 \text{ and } M34 = M3 \oplus M4$$

Step 7 : Complement the both M12 and M34 and append it together produces N= M12 || M34.

Step 8 : Map the appended binary digit into DNA strands using the Table 2.

Step 9 : Reduce the DNA strands by employing DNA-XOR Table 3 and produce the final DNA code.

TABLE I. CONVERSION OF PLAINTEXT INTO ASCII CODE

Alphabet	ASCII Code
A	65
B	67
C	68
D	69
E	70
F	71

TABLE II. MAP THE BINARY DIGIT INTO DNA STRANDS

Binary Digit	DNA strands
00	A
01	T
10	G
11	C

TABLE III. DNA-XOR

DNA-XOR	A	T	G	C
A	C	G	T	A
T	G	C	A	T
G	A	T	G	C
C	G	C	A	T

A. Internal Error Control Code based Authentication Algorithm

1) *Encryption Technique*: In internal error control code-based authentication algorithm, a novel DNA-XoR operation-based DNA code $D(m)$ is used as error control bit for authentication process to strengthen the security level. Initially, plaintext (m) converted into plaintext point by employing the koblitz method and resultant plaintext point P_m is directly affixed with generated DNA code and fed as input to ECC encryption algorithm as shown in Fig.1. Private Key (KA) of the user A encrypts the DNA coded message

input and produce the output as $E(KA [(m||D(m))])$. The encrypted message is allowed to pass through ECC decryption algorithm in which public key of user A involves to obtain the original DNA coded message.

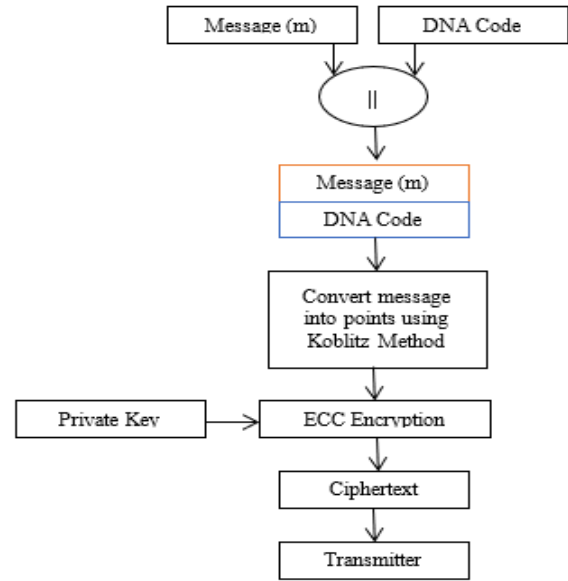


Fig. 1. Internal error control coding-based authentication algorithm – Transmitter side

2) *Decryption Technique*: At the receiver side, the DNA code and original message is detached from the output message as shown in Fig.2. The detached original message is served as input to novel DNA generation process to create a DNA code and it will be compared with received DNA code. If the calculated DNA code is equal to the incoming DNA code, then the received message is considered authentic.

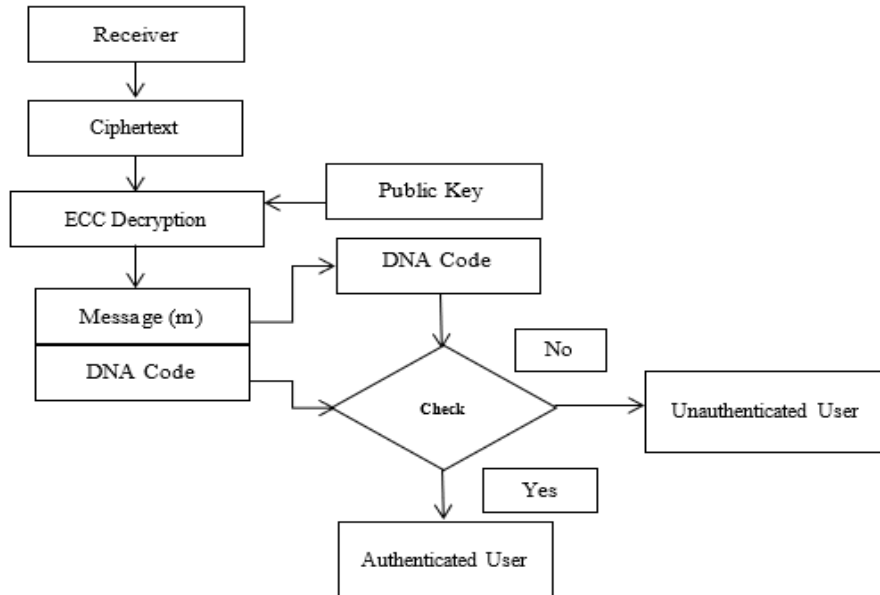


Fig. 2. Internal error control coding-based authentication algorithm – Receiver side

IV. RESULTS AND ITS DISCUSSION

The proposed internal error control coding based authentication algorithm consists of two techniques such as ECC encryption and DNA code generation process. Here,

DNA code generation process plays vital role for authentication process and level of security is improved by involving ECC techniques. MATLAB tool is used to implement the ECC algorithm and DNA code generation

algorithm and proves that it will be more suitable for IoT communication system. The proposed authentication algorithm is implemented and analyzed using different simulation parameters such as key size, data size, code length and processing time. Figure 3 & 4 shows the total running time of ECC encryption process and decryption process with respect to key length and compared with the traditional RSA algorithm. From the results, it prove that, ECC acquire less processing time compared with RSA algorithm and require lesser key length as shown in Table 4.

TABLE IV. COMPARISON OF RSA AND ECC WITH RESPECT TO DATA SIZE AND PROCESSING TIME

Data Security (in Bits)	RSA Key Length Required	Processing time (in ms)	ECC Key Length Required	Processing time (in ms)
80	1024	8.3	160	5.4
112	2048	14.6	224	8.6
128	3072	134.2	383	89.7
192	7680	535.87	511	389.54
256	15360	3486	1012	2678

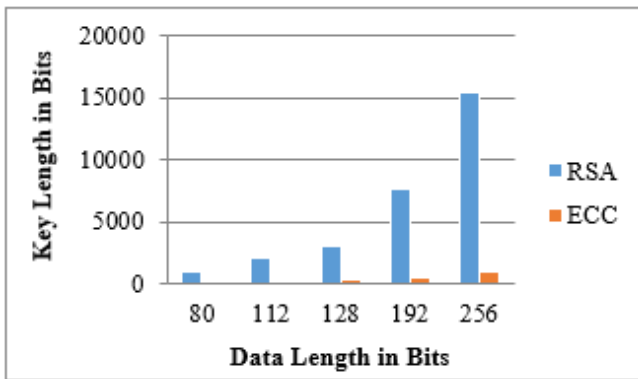


Fig. 3. Comparison of RSA & ECC with respect to key length

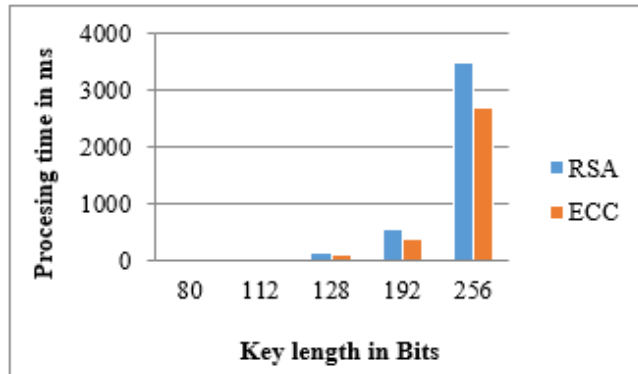


Fig. 4. Comparison of RSA & ECC with respect to processing time

TABLE V. TIME TAKEN TO GENERATE DNA CODE

Data Security (in Bits)	DNA Code generation processing time (in ms)
256	34
512	95
1024	172
2048	380
4096	560

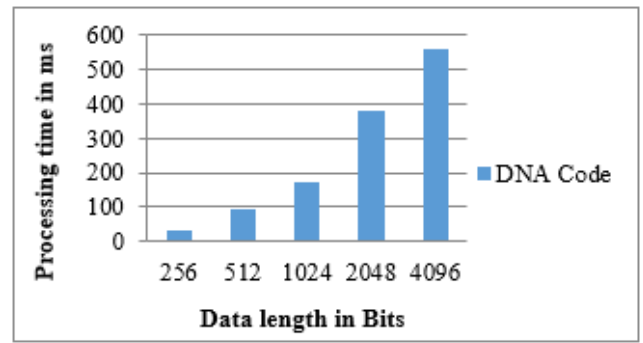


Fig. 5. Time taken to generate DNA code

Figure 5 & Table 5 give the total time taken to generate the DNA code for different set of data size such as 256, 512, 1024, 2048 and 4096. It proves that encoding 512 bits of data using DNA acquires lesser processing time. Figure 6 & Table 6 shows the total time taken to authenticate the user at the receiver side by checking the decrypted DNA code. It also shows the total time taken by the transmitter to encode the data using DNA mapping method and encryption process.

TABLE VI. TOTAL TIME TAKEN TO AUTHENTICATE THE USER

Data Security (in Bits)	Processing time (in ms)	
	Transmitter side	Receiver Side
256	2345	4567
512	6456	8985
1024	9456	10453
2048	29344	35056
4096	45764	56090

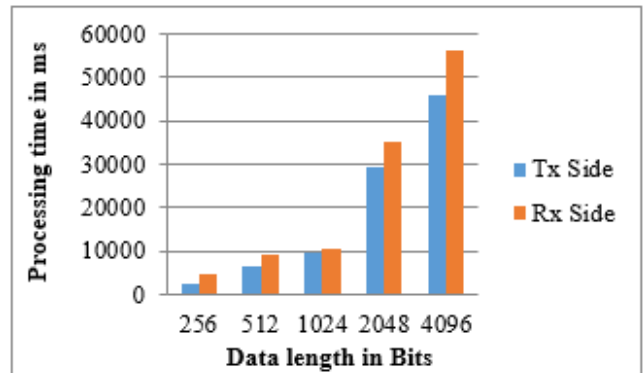


Fig. 6. Total time taken to authenticate the user

A. Discussion

The proposed algorithm is more suitable for IoT communication systems and is analyzed using various simulation parameters such as key size, data size, code length, and processing time. The results show that ECC acquires less processing time and requires lesser key length compared to the traditional RSA algorithm.

V. CONCLUSION

This paper presents a novel approach to securing patient data in the Medical Internet of Things (IoT) environment. The research addresses the limitations of traditional RSA cryptographic algorithms by introducing a lightweight Elliptic Curve Cryptography (ECC)-based authentication system that is specifically designed for resource-constrained IoT communication. Initially, importance of safeguarding patient identity in the modern medical field is presented to prevent

fraudulent use of personal information. Also, it mentioned about the use of biological features, such as DNA strands, for storing and sharing large amounts of patient data in encrypted form to enhance security. ECC-based authentication and integrity checking architecture with internal error control coding techniques to strengthen the authentication process is developed and incorporation of DNA cryptography and ECC encryption to achieve greater security for authenticating IoT devices while minimizing resource consumption. Future prospects for enhancing the proposed authentication system through double encryption to further bolster security in medical IoT communication systems. It highlights the significance of leveraging ECC and biological features for secure authentication in IoT communication, emphasizing the need for efficient and robust security measures in the healthcare sector.

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